

STAT6018 Research Frontiers in Data Science

Topic II: Introduction to empirical process theory

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References

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1 Chapter 1: Introduction to empirical processes

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What is an empirical process?

- A *stochastic process* is a collection of random variables $\{X(t), t \in T\}$ on the same probability space, indexed by an arbitrary index set T .
- In general, an *empirical process* is a stochastic process based on a random sample, usually of n i.i.d. random variables $X_1, \dots, X_n$.

Example: empirical distribution function

Let $X_1, \dots, X_n$ be i.i.d. real-valued random variables with cumulative distribution function (c.d.f.) F . Then the *empirical distribution function* (e.d.f.) is defined as

$$\mathbb{F}_n(t) := \frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_i \leq t), \quad t \in \mathbb{R}.$$

$\mathbb{F}_n(t)$ is one of the simplest examples of an empirical process.

Example: Kaplan-Meier estimator

Let $(X_1, \delta_1), \dots, (X_n, \delta_n)$ be a sample of right-censored failure time observations. Then the *Kaplan-Meier estimator* of the survival function is given by

$$\hat{S}(t) = \prod_{k: T_k^0 \leq t} \left\{ 1 - \frac{\sum_{i=1}^n \delta_i \mathbf{1}(X_i = T_k^0)}{\sum_{i=1}^n \mathbf{1}(X_i \geq T_k^0)} \right\},$$

where $T_1^0 < T_2^0 < \dots < T_K^0$ are unique observed failure times.

$\hat{S}(t)$ is another simple example of an empirical process.

General features of an empirical process

- The i.i.d. sample $X_1, \dots, X_n$ is drawn from a probability measure P on an arbitrary sample space $\mathcal{X}$.
- Define the *empirical measure* to be $\mathbb{P}_n = n^{-1} \sum_{i=1}^n \delta_{X_i}$, where δ_x denotes the Dirac measure at x .
- For a measurable function $f : \mathcal{X} \mapsto \mathbb{R}$, define

$$\mathbb{P}_n f := \int f d\mathbb{P}_n = \frac{1}{n} \sum_{i=1}^n f(X_i).$$

- For any class $\mathcal{F}$ of such real-valued functions on $\mathcal{X}$, $\{\mathbb{P}_n f : f \in \mathcal{F}\}$ is the empirical process indexed by $\mathcal{F}$.

Start with the classical e.d.f. $\mathbb{F}_n$

- Setting $\mathcal{X} = \mathbb{R}$, $\mathbb{F}_n$ can be re-expressed as the empirical process $\{\mathbb{P}_n f : f \in \mathcal{F}\}$, where $\mathcal{F} = \{\mathbb{1}(x \leq t), t \in \mathbb{R}\}$.
- By the law of large numbers, $\mathbb{F}_n(t) \xrightarrow{a.s.} F(t)$ for each $t \in \mathbb{R}$.
- By the central limit theorem, for each $t \in \mathbb{R}$,

$$\mathbb{G}_n(t) := \sqrt{n}(\mathbb{F}_n(t) - F(t)) \xrightarrow{d} N(0, F(t)(1 - F(t))).$$

- From the functional perspective, **uniform** results over $t \in \mathbb{R}$ would be more appealing.
 - ▶ **Need theory of empirical processes**

Strengthened results on $\mathbb{F}_n$ and $\mathbb{G}_n$

- Glivenko (1933) and Cantelli (1933) demonstrated that the previous result could be strengthened to

$$\|\mathbb{F}_n - F\|_\infty = \sup_{t \in \mathbb{R}} |\mathbb{F}_n(t) - F(t)| \xrightarrow{a.s.} 0.$$

- Donsker (1952) showed that

$$\mathbb{G}_n \xrightarrow{d} \mathbb{B}(F) \quad \text{in } \ell^\infty(\mathbb{R}),$$

where $\mathbb{B}$ is the standard Brownian bridge process on $[0, 1]$; for any index set T , $\ell^\infty(T)$ denotes the space of all bounded functions $f : T \mapsto \mathbb{R}$.

Extend to general empirical processes

- Properties of the approximation of Pf by $\mathbb{P}_n f$, **uniformly** in $\mathcal{F}$
 - ▶ the random quantity $\|\mathbb{P}_n - P\|_{\mathcal{F}} := \sup_{f \in \mathcal{F}} |\mathbb{P}_n f - Pf|$
 - ▶ the empirical process $\mathbb{G}_n := \sqrt{n}(\mathbb{P}_n - P)$

- Two special classes

- ▶ **Glivenko-Cantelli:** $\mathcal{F}$ is P -Glivenko-Cantelli if

$$\|\mathbb{P}_n - P\|_{\mathcal{F}} \xrightarrow{a.s.} 0.$$

- ▶ **Donsker:** $\mathcal{F}$ is P -Donsker if

$$\mathbb{G}_n \xrightarrow{d} \mathbb{G} \quad \text{in } \ell^\infty(\mathcal{F}),$$

where $\mathbb{G}$ is a mean zero Gaussian process indexed by $\mathcal{F}$, and $\ell^\infty(\mathcal{F}) = \{x : \mathcal{F} \mapsto \mathbb{R} \mid \|x\|_{\mathcal{F}} < \infty\}$.

Remarks

- Glivenko-Cantelli (GC): uniform almost surely convergence
- Donsker: uniform central limit theorem
- Donsker $\Rightarrow$ GC
- GC or Donsker properties depend crucially on the **complexity** of $\mathcal{F}$.

Complexity of $\mathcal{F}$

For a given norm $\|\cdot\|$, such as the $L_r(Q)$ -norms, define the covering and bracketing numbers as follows:

Covering number

- denoted by $N(\epsilon, \mathcal{F}, \|\cdot\|)$
- minimum number of balls $B(f; \epsilon) := \{g : \|g - f\| \leq \epsilon\}$ needed to cover $\mathcal{F}$
- *entropy without bracketing*: $\log N(\epsilon, \mathcal{F}, \|\cdot\|)$

Bracketing number

- denoted by $N_{[]}(\epsilon, \mathcal{F}, \|\cdot\|)$
- minimum number of brackets $[\ell, u]$ with $\|\ell - u\| < \epsilon$ needed to cover $\mathcal{F}$
- *entropy with bracketing*: $\log N_{[]}(\epsilon, \mathcal{F}, \|\cdot\|)$

GC theorems

Theorem 1 (GC with bracketing)

A function class $\mathcal{F}$ is a P -Glivenko-Cantelli if

$$N_{[]}(\epsilon, \mathcal{F}, L_1(P)) < \infty, \quad \text{for every } \epsilon > 0.$$

Theorem 2 (GC without bracketing)

A function class $\mathcal{F}$ is a P -Glivenko-Cantelli if

$$\sup_Q N(\epsilon \|F\|_{L_1(Q)}, \mathcal{F}, L_1(Q)) < \infty, \quad \text{for every } \epsilon > 0,$$

where F is an *envelope function*^a of $\mathcal{F}$, and the supremum is over all probability measures Q on $\mathcal{X}$.

^aAn envelope function of a class $\mathcal{F}$ is any function $x \mapsto F(x)$ such that $|f(x)| \leq F(x)$, for every x and $f \in \mathcal{F}$.

Donsker theorems

Theorem 3 (Donsker with bracketing entropy integral)

A function class $\mathcal{F}$ is a P -Donsker if

$$\int_0^\infty \sqrt{\log N_{[]}(\epsilon, \mathcal{F}, L_2(P))} d\epsilon < \infty.$$

Theorem 4 (Donsker with uniform entropy integral)

A function class $\mathcal{F}$ is a P -Donsker if

$$\int_0^\infty \sup_Q \sqrt{\log N(\epsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q))} d\epsilon < \infty,$$

where F is an envelope function of $\mathcal{F}$, and the supremum is over all probability measures Q on $\mathcal{X}$.

M-estimators

- Definition:

- ▶ Metric space: (Θ, d)
- ▶ $m_\theta : \mathcal{X} \rightarrow \mathbb{R}$, for each $\theta \in \Theta$
- ▶ “Empirical gain”: $M_n(\theta) = \mathbb{P}_n m_\theta$
- ▶ M-estimator: $\hat{\theta}_n = \arg \max_{\theta \in \Theta} M_n(\theta)$

- Examples:

- ▶ Maximum (penalized) likelihood estimator
- ▶ Least squares estimator
- ▶ Nonparametric maximum likelihood estimator

Application: consistency of M -estimators

- Two assumptions:

1. $\mathcal{F} := \{m_\theta(\cdot) : \theta \in \Theta\}$ is P -GC
2. θ_0 is a well-separated maximizer of $M(\theta) = Pm_\theta$, i.e., for every $\delta > 0$,
 $M(\theta_0) > \sup_{\theta \in \Theta: d(\theta, \theta_0) \geq \delta} M(\theta)$.

- For fixed $\delta > 0$, let $\psi(\delta) = M(\theta_0) - \sup_{\theta \in \Theta: d(\theta, \theta_0) \geq \delta} M(\theta) > 0$

$$\begin{aligned}\left\{d(\hat{\theta}_n, \theta_0) \geq \delta\right\} &\Rightarrow M(\hat{\theta}_n) \leq \sup_{\theta \in \Theta: d(\theta, \theta_0) \geq \delta} M(\theta) \\&\Leftrightarrow M(\hat{\theta}_n) - M(\theta_0) \leq -\psi(\delta) \\&\Rightarrow M(\hat{\theta}_n) - M(\theta_0) + \left(M_n(\theta_0) - M_n(\hat{\theta}_n)\right) \leq -\psi(\delta) \\&\Rightarrow 2 \sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \geq \psi(\delta)\end{aligned}$$

$$\Rightarrow \mathbb{P}\left(d(\hat{\theta}_n, \theta_0) \geq \delta\right) \leq \mathbb{P}\left(\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \geq \psi(\delta)/2\right) \rightarrow 0.$$

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Covering and packing numbers

Let (Θ, d) be an arbitrary semi-metric space.

Definition 5 (Covering number)

The ϵ -covering number $N(\epsilon, \Theta, d)$ is the minimal number of balls $B(x; \epsilon) := \{y \in \Theta : d(x, y) \leq \epsilon\}$ of radius ϵ needed to cover the set Θ . The corresponding entropy number is $\log N(\epsilon, \Theta, d)$.

Definition 6 (Packing number)

Call a collection of points ϵ -separated if the distance between each pair of points is larger than ϵ . The packing number $D(\epsilon, \Theta, d)$ is the maximum number of ϵ -separated points in Θ .

Covering and packing numbers (cont.)

Lemma 7 (Covering vs packing numbers)

$$D(2\epsilon, \Theta, d) \leq N(\epsilon, \Theta, d) \leq D(\epsilon, \Theta, d), \quad \forall \epsilon > 0.$$

Thus, packing and covering numbers have the same scaling in the radius ϵ .

- The first inequality can be easily proved by contradiction.
- The second inequality follows by the fact that Θ can be covered by the balls $B(\theta_i; \epsilon)$ ($i = 1, \dots, D$), where $\theta_1, \dots, \theta_D$ are the ϵ -separated points associated with the packing number D .

Example: bounded sets on Euclidean space

Example 8 (Bounded sets on Euclidean space)

For any bounded subset $\Theta \subset \mathbb{R}^p$, there exist constants $c < C$ such that

$$c \left(\frac{1}{\epsilon} \right)^p \leq N(\epsilon, \Theta, \|\cdot\|) \leq C \left(\frac{1}{\epsilon} \right)^p, \quad \forall \epsilon \in (0, 1).$$

Proof.

The union of $D(\epsilon, \Theta, \|\cdot\|)$ number of ϵ -separated balls of radius $\epsilon/2$ is contained in the set $\Theta' := \{\theta \in \mathbb{R}^p : \|\theta - \Theta\| < \epsilon/2\}$. Thus, $D(\epsilon, \Theta, \|\cdot\|) v_p \left(\frac{\epsilon}{2} \right)^p \leq \text{Vol}(\Theta')$, where v_p is the volume of the unit ball. On the other hand, $D(2\epsilon, \Theta, \|\cdot\|)$ number of 2ϵ -separated balls cover the set Θ . Thus, $D(2\epsilon, \Theta, \|\cdot\|) v_p (2\epsilon)^p \geq \text{Vol}(\Theta)$. The desired inequalities then follow by the above results and Lemma 7. □

Example: bounded Lipschitz functions

Example 9 (Bounded Lipschitz functions)

Let $\mathcal{F} := \{f : [0, 1] \mapsto [0, 1] \mid f \text{ is 1-Lipschitz}\}$. Then there exists some constant A such that

$$\log N(\epsilon, \mathcal{F}, \|\cdot\|_\infty) \leq \frac{A}{\epsilon}, \quad \forall \epsilon > 0.$$

Proof.

If $\epsilon \geq 1$, take $f_0 \equiv 0$ and observe that $\forall f \in \mathcal{F}$, $\|f - f_0\|_\infty \leq 1 \leq \epsilon$. Then $N(\epsilon, \mathcal{F}, \|\cdot\|_\infty) = 1$.

Let $0 < \epsilon < 1$. Define a ϵ -grid of the interval $[0, 1]$ (for both axes), i.e. $0 = a_0 < a_1 < \dots, a_N = 1$ where $N = \lfloor 1/\epsilon \rfloor + 1$ and $a_k = k\epsilon$ for $k = 1, \dots, N-1$.

Let $B_1 := [a_0, a_1]$ and $B_k := (a_{k-1}, a_k]$ for $k = 2, \dots, N$.

Example: bounded Lipschitz functions (cont.)

Proof (cont.)

For each $f \in \mathcal{F}$, define the step function $\tilde{f} : [0, 1] \mapsto \mathbb{R}$ as

$$\tilde{f}(x) = \sum_{k=1}^N \epsilon \left\lfloor \frac{f(a_k)}{\epsilon} \right\rfloor \mathbb{1}_{B_k}(x).$$

Clearly, $\tilde{f}$ is constant on each interval B_k and can only take values of the form $i\epsilon$ for $i = 0, \dots, N-1$.

For any $x \in [0, 1]$, suppose that $x \in B_k$. By the Lipschitz property of f and the construction of $\tilde{f}$,

$$|f(x) - \tilde{f}(x)| \leq |f(x) - f(a_k)| + |f(a_k) - \tilde{f}(a_k)| \leq 2\epsilon.$$

Therefore, $\|f - \tilde{f}\|_{\infty} \leq 2\epsilon$.

Example: bounded Lipschitz functions (cont.)

Proof (cont.)

Now we count the number of distinct $\tilde{f}$'s obtained as f varies over $\mathcal{F}$. There are at most N choices for $\tilde{f}(a_1)$. Further, note that for any $\tilde{f}$ and $k = 2, \dots, N$,

$$\begin{aligned} & |\tilde{f}(a_k) - \tilde{f}(a_{k-1})| \\ & \leq |\tilde{f}(a_k) - f(a_k)| + |f(a_k) - f(a_{k-1})| + |f(a_{k-1}) - \tilde{f}(a_{k-1})| \leq 3\epsilon. \end{aligned}$$

Thus, for fixed $\tilde{f}(a_{k-1})$, there are at most 7 choices left for $\tilde{f}(a_k)$. Therefore,

$$N(2\epsilon, \mathcal{F}, \|\cdot\|_\infty) \leq (\lfloor 1/\epsilon \rfloor + 1) 7^{\lfloor 1/\epsilon \rfloor},$$

which completes the proof. □

Bracketing numbers

Let $(\mathcal{F}, \|\cdot\|)$ be a subset of a normed space of real functions $f : \mathcal{X} \mapsto \mathbb{R}$ on some set $\mathcal{X}$.

Definition 10 (Bracketing number)

Given two functions $l(\cdot)$ and $u(\cdot)$, the bracket $[l, u]$ is the set of all functions $f \in \mathcal{F}$ with $l(x) \leq f(x) \leq u(x), \forall x \in \mathcal{X}$. An ϵ -bracket is a bracket $[l, u]$ with $\|l - u\| < \epsilon$. The bracketing number $N_{[]}(\epsilon, \mathcal{F}, \|\cdot\|)$ is the minimum number of the ϵ -brackets needed to cover $\mathcal{F}$. The entropy with bracketing is $\log N_{[]}(\epsilon, \mathcal{F}, \|\cdot\|)$.

Bracketing numbers (cont.)

Theorem 11 (Bracketing vs covering numbers)

Suppose that $\|\cdot\|$ has the Riesz property^a. Then

$$N(\epsilon, \mathcal{F}, \|\cdot\|) \leq N_{[]} (2\epsilon, \mathcal{F}, \|\cdot\|), \quad \forall \epsilon > 0.$$

^a $|f| \leq |g|$ implies that $\|f\| \leq \|g\|$.

- The proof uses the fact that every f within the 2ϵ -bracket $[l, u]$ falls within the ball $B(\frac{l+u}{2}; \epsilon)$.
- In general, there is no converse inequality, so that bracketing numbers are bigger than covering numbers.
- A bracket gives pointwise control over a function.
- A ball under the $L_r(Q)$ -norm gives integrated control over a function.

Example: distribution functions

Example 12 (Distribution functions)

Recall that the function class relevant to the e.d.f. $\mathbb{F}_n$ is $\mathcal{F} = \{\mathbb{1}_{(-\infty, t]} \mid t \in \mathbb{R}\}$. The bracketing numbers of $\mathcal{F}$ are of polynomial orders:

$$N_{[]}(\epsilon, \mathcal{F}, L_1(P)) \leq \frac{2}{\epsilon},$$
$$N_{[]}(\epsilon, \mathcal{F}, L_2(P)) \leq \frac{2}{\epsilon^2}.$$

Proof.

Consider the brackets of the form $[\mathbb{1}_{(-\infty, t_{i-1}]}, \mathbb{1}_{(-\infty, t_i]}]$ for a grid of points $-\infty = t_0 < t_1 < \dots < t_N = \infty$ such that $F(t_i) - F(t_{i-1}) < \epsilon$ for $i = 1, \dots, N$, where $N = \lfloor 1/\epsilon \rfloor + 1 < 2/\epsilon$.

Clearly, these brackets can cover $\mathcal{F}$. Moreover, these brackets have $L_1(P)$ -size ϵ and $L_2(P)$ -size bounded by $\sqrt{\epsilon}$ (since $Pf^2 \leq Pf$ for every $0 \leq f \leq 1$). $\square$

Example: classes Lipschitz in a parameter

Example 13 (Classes Lipschitz in a parameter)

Consider a function class $\mathcal{F} = \{m_\theta : \theta \in \Theta\}$ which has a Lipschitz dependence on θ , i.e., there exists some function $F : \mathcal{X} \mapsto \mathbb{R}$ such that

$$|m_{\theta_1}(x) - m_{\theta_2}(x)| \leq F(x)d(\theta_1, \theta_2), \quad \forall \theta_1, \theta_2 \in \Theta, \forall x \in \mathcal{X}.$$

Then, for any norm $\|\cdot\|$,

$$N_{[]} (2\epsilon \|F\|, \mathcal{F}, \|\cdot\|) \leq N(\epsilon, \Theta, d).$$

Proof.

Let $\theta_1, \dots, \theta_p$ be an ϵ -cover of Θ (under the metric d).

Then for every $\theta \in B(\theta_i; \epsilon)$, $|m_\theta(x) - m_{\theta_i}(x)| \leq \epsilon F(x)$.

Thus, the brackets $[m_{\theta_i} - \epsilon F, m_{\theta_i} + \epsilon F]$ ($i = 1, \dots, p$), each of size $2\epsilon \|F\|$, can cover $\mathcal{F}$. □

Monotone functions

Theorem 14 (Monotone functions)

The class $\mathcal{F}$ of monotone functions $f : \mathbb{R} \mapsto [0, 1]$ satisfies

$$\log N_{[]}(\epsilon, \mathcal{F}, L_r(Q)) \leq K\left(\frac{1}{\epsilon}\right), \quad \forall \epsilon > 0,$$

for every probability measure Q , every $r \geq 1$, and some constant K that depends on r only.

- The result implies that $\mathcal{F}$ is Donsker (by Theorem 3).
- See Theorem 2.7.5 of VW for the proof.

Smooth functions

- $\mathcal{X}$: bounded, convex subset of $\mathbb{R}^p$ with nonempty interior
- $\underline{\alpha}$: largest integer smaller than α , for any $\alpha > 0$
- D^k : differential operator of order k
- For a function $f : \mathcal{X} \mapsto \mathbb{R}$, define

$$\|f\|_{\alpha} = \max_{k \leq \underline{\alpha}} \sup_{D^k, x} |D^k f(x)| + \sup_{D^{\alpha}, x, y} \frac{|D^{\alpha} f(x) - D^{\alpha} f(y)|}{\|x - y\|^{\alpha - \underline{\alpha}}}$$

- $C_M^{\alpha}(\mathcal{X})$: set of all continuous functions $f : \mathcal{X} \mapsto \mathbb{R}$ with $\|f\|_{\alpha} \leq M$ (f has uniformly bounded partial derivatives and the highest partial derivatives are Lipschitz)

Smooth functions (cont.)

Theorem 15 (Smooth functions)

There exists a constant K depending only on α , $\text{diam}\mathcal{X}$, and p such that

$$\log N(\epsilon, C_1^\alpha(\mathcal{X}), \|\cdot\|_\infty) \leq K \left(\frac{1}{\epsilon}\right)^{p/\alpha},$$
$$\log N_{[]}(\epsilon, C_1^\alpha(\mathcal{X}), L_r(Q)) \leq K \left(\frac{1}{\epsilon}\right)^{p/\alpha},$$

for every $\epsilon > 0$, $r \geq 1$, and probability measure Q .

See Theorem 2.7.1 and Corollary 2.7.2 of VW for the proofs.

Convex functions

Theorem 16 (Convex functions)

For a compact, convex subset $C \subset \mathbb{R}^p$, the class $\mathcal{F}$ of all convex functions $f : C \mapsto [0, 1]$ that are L -Lipschitz satisfies

$$\log N(\epsilon, \mathcal{F}, \|\cdot\|_\infty) \leq K(1 + L)^{p/2} \left(\frac{1}{\epsilon}\right)^{p/2},$$

for some constant K depending on p and C only.

See Corollary 2.7.10 of VW for the proof.

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Tail probability of random variables

- **Markov's inequality**

Let $Z \geq 0$ be a random variable. Then for any $t > 0$,

$$P(Z \geq t) \leq \frac{EZ}{t}.$$

- **Chebyshev's inequality**

If Z has a finite variance $\text{Var}(Z)$, then

$$P(|Z - EZ| \geq t) \leq \frac{\text{Var}(Z)}{t^2}.$$

But these inequalities can only yield a tail bound of order t^{-2} , which may be too relaxed. The tail bound can be improved to an exponential decrease in t^2 by Hoeffding's inequality.

Hoeffding's inequality

Lemma 17 (Hoeffding's inequality)

Let $X_1, \dots, X_n$ be independent bounded random variables such that $X_i \in [a_i, b_i]$ with probability 1. Let $S_n = \sum_{i=1}^n X_i$. Then,

$$P(S_n - ES_n \geq t) \leq e^{-2t^2 / \sum_{i=1}^n (b_i - a_i)^2},$$
$$P(S_n - ES_n \leq -t) \leq e^{-2t^2 / \sum_{i=1}^n (b_i - a_i)^2}.$$

The proof uses Markov's inequality and the following lemma:

Lemma 18

Let X be a random variable with $EX = 0$ and $X \in [a, b]$ with probability 1. Then for any $\lambda > 0$,

$$E(e^{\lambda X}) \leq e^{\lambda^2(b-a)^2/8}.$$

Sub-Gaussian random variables

Definition 19 (Sub-Gaussian random variables)

A random variable X is called *sub-Gaussian* if there exist constants $C, v > 0$ such that $P(|X| > t) \leq Ce^{-vt^2}$ for every $t > 0$.

Some equivalent characterizations of sub-Gaussian random variables:

- There exists $a > 0$ such that $E[e^{aX^2}] < \infty$.
- Laplace transform condition: $\exists B, b > 0$ such that $\forall \lambda \in \mathbb{R}, Ee^{\lambda(X-E[X])} \leq Be^{\lambda^2 b}$.
- Moment condition: $\exists K > 0$ such that $\forall p \geq 1, (E|X|^p)^{1/p} \leq K\sqrt{p}$.
- Union bound condition: $\exists c > 0$ such that $\forall n \geq c,$

$$E[\max\{|X_1 - E[X]|, \dots, |X_n - E[X]|\}] \leq c\sqrt{\log n}$$

where $X_1, \dots, X_n$ are i.i.d. copies of X .

Sub-Gaussian processes

Definition 20 (Sub-Gaussian processes)

Let (T, d) be a semi-metric space and $\{X_t, t \in T\}$ be a stochastic process indexed by T . Then X_t is called sub-Gaussian w.r.t. the semi-metric d if

$$P(|X_s - X_t| > u) \leq 2 \exp\left(-\frac{u^2}{2d(s, t)^2}\right), \quad \forall s, t \in T, u > 0.$$

Any Gaussian process is sub-Gaussian w.r.t. the standard deviation semi-metric $d(s, t) = \sqrt{\text{Var}(X_s - X_t)}$.

Rademacher process and Hoeffding's inequality

Consider the *Rademacher process*

$$X_a = \sum_{i=1}^n a_i \varepsilon_i, \quad a = (a_1, \dots, a_n) \in \mathbb{R}^n, \quad (1)$$

where ε_i 's are independent Rademacher variables which take values $+1$ and -1 with probability $1/2$.

By the following special case of Hoeffding's inequality, Rademacher process is also sub-Gaussian (w.r.t. the Euclidean distance).

Lemma 21 (Hoeffding's inequality)

The Rademacher process $\{X_a : a \in \mathbb{R}^n\}$ defined in (1) satisfies

$$P(|X_a| > t) \leq 2e^{-t^2/(2\|a\|^2)}.$$

Bernstein's inequality

The following result gives tail bounds for random variables with larger than normal tails.

Lemma 22 (Bernstein's inequality)

For independent random variables $Y_1, \dots, Y_n$ with zero means and bounded ranges $[-M, M]$, there exists a constant $v \geq \text{Var}(\sum_{i=1}^n Y_i)$ such that

$$P(|\sum_{i=1}^n Y_i| > t) \leq 2e^{-\frac{t^2}{2(v+Mt/3)}}.$$

- See page 855 of Shorack and Wellner (1986)¹ for the proof.
- Compared to the normal tail bound $e^{-t^2/(2v)}$, the extra term $2Mt/3$ can be seen as a penalty for the non-normality.
- When $n \rightarrow \infty$, $Mt/3$ is typically negligible w.r.t. v .

¹Shorack, G. R., & Wellner, J. A. (1986). *Empirical Processes with Applications to Statistics*. Wiley, New York.

Maximal inequalities

Lemma 23 (Maximal inequality for sub-Gaussian variables)

Suppose that $Y_1, \dots, Y_N$ (not necessarily independent) are sub-Gaussian in the sense that $Ee^{\lambda Y_i} \leq e^{\lambda^2 \sigma^2 / 2}$ for all $\lambda > 0$ and $i = 1, \dots, N$. Then,

$$E \max_{i=1, \dots, N} Y_i \leq \sigma \sqrt{2 \log N}.$$

Proof.

By Jensen's inequality, we have

$$e^{\lambda E \max_{i=1, \dots, N} Y_i} \leq E e^{\lambda \max_{i=1, \dots, N} Y_i} \leq \sum_{i=1}^N E e^{\lambda Y_i} \leq N e^{\lambda^2 \sigma^2 / 2}.$$

Taking logarithms yields

$$E \max_{i=1, \dots, N} Y_i \leq \frac{\log N}{\lambda} + \frac{\lambda \sigma^2}{2} \leq \sigma \sqrt{2 \log N}.$$



Maximal inequalities (cont.)

Lemma 24

Let ψ be a strictly increasing, convex, non-negative function. Suppose that $\xi_1, \dots, \xi_N$ are random variables such that $E[\psi(|\xi_i|/c_i)] \leq L$ for $i = 1, \dots, N$ and some constant L . Then,

$$E \max_{1 \leq i \leq N} |\xi_i| \leq \psi^{-1}(LN) \max_{1 \leq i \leq N} c_i.$$

Proof.

By the properties of ψ ,

$$\psi \left(\frac{E \max_{1 \leq i \leq N} |\xi_i|}{\max_{1 \leq i \leq N} c_i} \right) \leq \psi \left(E \max_{1 \leq i \leq N} \frac{|\xi_i|}{c_i} \right) \leq \sum_{i=1}^N E \psi \left(\frac{|\xi_i|}{c_i} \right) \leq LN.$$

Apply ψ^{-1} to both sides.



Maximal inequalities (cont.)

Corollary 25

Let $\xi_1, \dots, \xi_N$ be Rademacher linear combinations, i.e., $\xi_i = \sum_{k=1}^n a_k^{(i)} \varepsilon_k$. Then there exists some constant $C > 0$ such that for $N \geq 2$,

$$E \max_{1 \leq i \leq N} |\xi_i| \leq C \sqrt{\log N} \max_{1 \leq i \leq N} \|a^{(i)}\|,$$

where $a^{(i)} = (a_1^{(i)}, \dots, a_n^{(i)}) \in \mathbb{R}^n$.

Proof.

Use the fact that $E[e^{\xi_i^2 / (6\|a^{(i)}\|^2)}] \leq 2$ and Lemma 24 with $\psi(x) = e^{x^2}$. □

Symmetrization

Symmetrized empirical process:

$$f \mapsto \mathbb{P}_n^o f = \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i),$$

where $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d. Rademacher random variables.

- $\varepsilon_1, \dots, \varepsilon_n$ are independent of $(X_1, \dots, X_n)$
- $E(\mathbb{P}_n^o f) = 0$
- For fixed $(X_1, \dots, X_n)$, $\mathbb{P}_n^o$ is a Rademacher process (hence sub-Gaussian).

Symmetrization result

Theorem 26 (Symmetrization)

For any class $\mathcal{F}$ of measurable functions,

$$E \|\mathbb{P}_n - P\|_{\mathcal{F}} \leq 2E \|\mathbb{P}_n^o\|_{\mathcal{F}}.$$

Proof.

Let Y_i be independent copies of X_i . For fixed $(X_1, \dots, X_n)$,

$$\|\mathbb{P}_n - P\|_{\mathcal{F}} = \sup_{f \in \mathcal{F}} \frac{1}{n} \left| \sum_{i=1}^n [f(X_i) - Ef(Y_i)] \right| \leq E_Y \sup_{f \in \mathcal{F}} \frac{1}{n} \left| \sum_{i=1}^n [f(X_i) - f(Y_i)] \right|.$$

Taking expectation with respect to $(X_1, \dots, X_n)$, we obtain

$$E \|\mathbb{P}_n - P\|_{\mathcal{F}} \leq E \left\| \frac{1}{n} \sum_{i=1}^n [f(X_i) - f(Y_i)] \right\|_{\mathcal{F}}.$$

Symmetrization result (cont.)

Proof (cont.)

We can see that adding a minus sign in front of $[f(X_i) - f(Y_i)]$ just exchanges X 's and Y 's, so the expectation remains unchanged. Thus,

$E \frac{1}{n} \left\| \sum_{i=1}^n \varepsilon_i [f(X_i) - f(Y_i)] \right\|_{\mathcal{F}}$ is the same for any $(\varepsilon_1, \dots, \varepsilon_n) \in \{-1, +1\}^n$.
Hence,

$$\begin{aligned} E \|\mathbb{P}_n - P\|_{\mathcal{F}} &\leq E_{\varepsilon} E_{X,Y} \left\| \frac{1}{n} \sum_{i=1}^n \varepsilon_i [f(X_i) - f(Y_i)] \right\|_{\mathcal{F}} \\ &\leq E_{\varepsilon} E_X \left\| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right\|_{\mathcal{F}} + E_{\varepsilon} E_Y \left\| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(Y_i) \right\|_{\mathcal{F}} \\ &= 2E \|\mathbb{P}_n^o\|_{\mathcal{F}}. \end{aligned}$$

